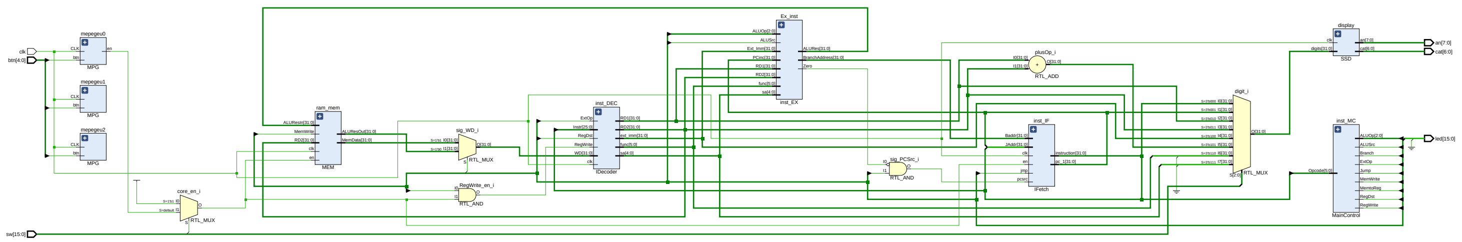
clk
btn[4:0]
sw[15:0]
mepegeu0
mepegeu1
mepegeu2
CLK
btn
en
MPG
core_en_i
S=1'b1 I0
S=default I1
O
S
RTL_MUX
ram_mem
ALUResIn[31:0]
MemWrite
RD2[31:0]
clk
en
ALUResOut[31:0]
MemData[31:0]
MEM
sig_WD_i
S=1'b1 I0[31:0]
S=1'b0 I1[31:0]
O[31:0]
S
RTL_MUX
RegWrite_en_i
I0
I1
O
RTL_AND
inst_DEC
ExtOp
Instr[25:0]
RegDst
RegWrite
WD[31:0]
clk
RD1[31:0]
RD2[31:0]
ext_imm[31:0]
func[5:0]
sa[4:0]
IDecoder
Ex_inst
ALUOp[2:0]
ALUSrc
Ext_Imm[31:0]
PCinc[31:0]
RD1[31:0]
RD2[31:0]
func[5:0]
sa[4:0]
ALURes[31:0]
BranchAddress[31:0]
Zero
inst_EX
sig_PCSrc_i
I0
I1
O
RTL_AND
plusOp_i
I0[31:0]
I1[31:0]
O[31:0]
+
RTL_ADD
inst_IF
Baddr[31:0]
JAddr[31:0]
clk
en
jmp
pcsrc
instruction[31:0]
pc_1[31:0]
IFetch
digit_i
S=3'b000 I0[31:0]
S=3'b001 I1[31:0]
S=3'b010 I2[31:0]
S=3'b011 I3[31:0]
S=3'b100 I4[31:0]
S=3'b101 I5[31:0]
S=3'b110 I6[31:0]
S=3'b111 I7[31:0]
O[31:0]
S[2:0]
RTL_MUX
display
clk
digits[31:0]
an[7:0]
cat[6:0]
SSD
an[7:0]
cat[6:0]
inst_MC
Opcode[5:0]
ALUOp[2:0]
ALUSrc
Branch
ExtOp
Jump
MemWrite
MemtoReg
RegDst
RegWrite
MainControl
led[15:0]